

Topics to be covered

- Transformations of the independent variable
- Systems, Properties of systems
- LTI systems, Sampling property, Convolution
- LCCDE, Impulse response, Transfer function
- Fourier series, Existence conditions, Gibbs phenomenon
- Properties of Fourier series

Continuous-time (C-T) signals are represented as $x(t)$.

Discrete-time (D-T) signals are represented as $x[n]$.

Useful Signal Measures

- **Energy -**

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt \quad (\text{C-T})$$

$$E_x = \sum_{n=-\infty}^{\infty} |x[n]|^2 \quad (\text{D-T})$$

- **Power -**

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T |x(t)|^2 dt \quad (\text{C-T})$$

$$P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2 \quad (\text{D-T})$$

Transformations of the Independent Variable

1) Time-Shifting

The delayed signal is $x_1(t) = x(t - t_0)$, $t_0 > 0$;
similarly the advanced signal is $x_2(t) = x(t + t_0)$, $t_0 > 0$.

2) Time-Scaling

The slowed-down signal is $x_1(t) = x(t/\alpha)$;
the sped-up signal is $x_2(t) = x(\alpha t)$, where $\alpha > 1$ is the scaling factor.

3) Time-Reversal The signal reflected about the y-axis is $x_1(t) = x(-t)$

Transformations of the Independent Variable

In general, a transformation may look like $x_*(t) = x(at + b)$

We can arrive at this transformation in two ways -

Step 1) Time-scaling by a , ie,

$$x_1(t) = x(at)$$

Step 2) Time-shift by b/a , ie,

$$x_2(t) = x_1\left(t + \frac{b}{a}\right) = x(at + b)$$

OR

Step 1) Time-shift by b , ie,

$$x_1(t) = x(t + b)$$

Step 2) Time-scaling by a , ie,

$$x_2(t) = x_1(at) = x(at + b)$$

Some remarks on D-T Complex Exponential Signals

$$x[n] = e^{j\omega_0 n}$$

Periodicity -

Discrete-time complex exponential signals are only periodic if -

$$e^{j\omega_0(n+N)} = e^{j\omega_0 n} \implies e^{j\omega_0 N} = 1$$

$$\therefore \omega_0 = 2\pi \frac{m}{N}, \quad m \in \mathbb{Z}$$

Only if this is satisfied, then the signal is periodic with period N
Otherwise it is not periodic. (Unlike continuous-time!)

Non-uniqueness -

$$e^{j\omega n} = e^{j(\omega+2\pi m)n} \quad \forall m \in \mathbb{Z}$$

The range of unique discrete-time frequencies is only $[0, 2\pi)$

A **system** takes in input signals, and produces output signals based on some rules of operation that describe its behaviour.

Denoted as -

$y(t) = T\{x(t)\}$ or simply $x(t) \rightarrow y(t)$ for continuous time systems

$y[n] = T\{x[n]\}$ or simply $x[n] \rightarrow y[n]$ for discrete time systems

Properties of Systems

Causal System

A system is called **causal** if the output at time t_0 (or n_0) depends only on the input for $t \leq t_0$ (or $n \leq n_0$).

Most systems in nature are causal.

E.g., $y(t) = (x(t))^2$ and $y(t) = \int_{-\infty}^t x(\tau) d\tau$ are causal;

$y(t) = x(t-1) + x(t+1)$ is **not** causal.

Memoryless System

A system is called **memoryless** if the output at time t_0 (or n_0) depends only on the input value at $t = t_0$ (or $n = n_0$). Memoryless systems are a special case of causal systems.

E.g., $y(t) = (x(t))^2$ is memoryless; $y(t) = \int_{-\infty}^t x(\tau) d\tau$ is **not** memoryless.

Invertible System

A system is called **invertible** if it produces distinct outputs for distinct inputs. For an invertible system, we can find an inverse system.

E.g., $y(t) = 2x(t)$ is invertible; $y(t) = (x(t))^2$ is **not** invertible.

Stable System

A system is called **BIBO stable** (bounded-input-bounded-output) if it produces bounded outputs for all bounded input signals.

A signal $x(t)$ is called bounded if $|x(t)| < M \forall t$, where $M < \infty$.

E.g., $y(t) = (x(t))^2$ is BIBO stable; $y(t) = \int_{-\infty}^t x(\tau) d\tau$ is **not** BIBO stable.

Properties of Systems

Time-Invariant System

A system is called **time-invariant** if a time-shift in the input signal produces the same time-shift in the output signal.

$$x(t) \rightarrow y(t) \quad \implies \quad x(t - \tau) \rightarrow y(t - \tau)$$

Linear System

A system is called **linear** if, given

$$x_1(t) \rightarrow y_1(t) \quad \text{and} \quad x_2(t) \rightarrow y_2(t),$$

then for any complex constants a and b ,

$$ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$$

Systems that are both linear and time-invariant are called **LTI systems**.

Impulse Function

Continuous-time - It assumes a very high value (tending to infinity) in a very narrow region (width tending to zero) around $t = 0$, and is zero everywhere else; such that the area under the function is 1. It is defined by the properties -

$$\delta(t) = 0 \quad \forall t \neq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t) dt = 1$$

Discrete-time - Here, the impulse function is well-defined, and can be written explicitly as -

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Sampling Property

Continuous-time -

$$x(t) = \int_{-\infty}^{\infty} x(\tau) \delta(t - \tau) d\tau$$

Discrete-time -

$$x[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - k]$$

Intuition - These results tell us that any signal $x(t)$ (or $x[n]$) can be represented as a *linear combination of time-shifted impulse functions*, $\delta(t - \tau)$ (or $\delta[n - k]$), with coefficients $x(\tau)$ (or $x[k]$).

LTI Systems and Impulse Response

The complete behaviour of an LTI system towards any general input is characterized by its **impulse response** - its output to an impulse function input.

Continuous-time - Let $h(t)$ be the impulse response of an LTI system. For any general input $x(t)$, the output is given by -

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = x(t) * h(t)$$

This is called the **convolution integral**.

Discrete-time - Let $h[n]$ be the impulse response of an LTI system. For any general input $x[n]$, the output is given by -

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k] = x[n] * h[n]$$

LTI System Properties using Impulse Response

Causality

An LTI system is **causal** iff its impulse response $h(t) = 0 \forall t < 0$, because then the convolution integral becomes independent of $x(t - \tau)$ for all future values of τ .

Memorylessness

An LTI system is **memoryless** iff $h(t) = 0 \forall t \neq 0$.

Hence, for a memoryless LTI system, $h(t) = k\delta(t)$, i.e. the **scalar-amplifier system**, $y(t) = kx(t)$ is the only memoryless LTI system, (Other memoryless systems exist, but they are not LTI, e.g. $y(t) = (x(t))^2$.)

Invertibility

An LTI system is **invertible** iff there exists a signal $h_i(t)$ such that $h(t) * h_i(t) = \delta(t)$

LTI System Properties using Impulse Response

Stability

An LTI system is **BIBO stable iff** its impulse response is **absolutely integrable** (absolutely summable in the discrete case). This is both a **necessary and sufficient** condition.

Continuous-time -

$$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$$

Discrete-time -

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

LCCDE Representation of LTI Systems

Systems are often represented in terms of a differential (or difference) equation relating the input and output.

Continuous-time - A Linear Constant-Coefficient Differential Equation (LCCDE) has the general form -

$$\sum_{k=0}^N a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^M b_k \frac{d^k x(t)}{dt^k}$$

Discrete-time - A Linear Constant-Coefficient Difference Equation (LCCDE) has the general form -

$$\sum_{k=0}^N a_k y[n-k] = \sum_{k=0}^M b_k x[n-k]$$

The highest order derivative (or delay) of the output, N , is called the **order** of the system.

Not every LTI system can be represented by an LCCDE - e.g. a pure time-delay, $y(t) = x(t - t_0)$, cannot be.

Linearity of Systems represented by LCCDE

To actually determine the output of an LCCDE for a given input, we need N **auxiliary conditions**.

A system described by an LCCDE need not necessarily be linear - this also depends on the auxiliary conditions specified.

For example, consider the d-t system,

$$y[n] - ay[n-1] = x[n]; \quad y[-1] = 1$$

Check that this is not linear!

We define the condition of **initial rest** as -

$$x(t) = 0 \quad \forall t < t_0 \quad \implies \quad y(t) = 0 \quad \forall t < t_0$$

A system described by an LCCDE and satisfying initial rest is guaranteed to be **LTI** (linear and causal).

Eigenfunctions of LTI Systems

Consider the output of an LTI system, with impulse response $h(t)$, to the complex exponential input $x(t) = e^{st}$, $s = \sigma + j\omega$,

$$y(t) = h(t) * e^{st} = \int_{-\infty}^{\infty} h(\tau) e^{s(t-\tau)} d\tau = e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

Defining $H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$, we get -

$$y(t) = H(s) e^{st}$$

The output to a complex exponential input is the **same** complex exponential, scaled only by the complex number $H(s)$.

- Hence, e^{st} is called an **eigenfunction** of the LTI system, and $H(s)$ the corresponding **eigenvalue**.
- $H(s)$ is called the **Transfer Function** (or **System Function**). It is an alternative way to characterize LTI systems.

Function Space

We can think of a signal $x(t)$ as a **vector** in an infinite-dimensional vector space - one "coordinate" for every value of t .

Just as with ordinary vectors, we can define an **inner product** between two signals (vectors) $x(t)$ and $y(t)$ -

$$\langle x(t), y(t) \rangle = \frac{1}{T} \int_T x(t) y^*(t) dt$$

where the integral is taken over the interval of interest (e.g. one period T , for periodic signals).

(Here, the factor of $1/T$ is included since we are considering periodic functions. For more general function space containing aperiodic functions, we may define the inner product without the factor of $1/T$)

Complex Exponentials as a Basis Set

Consider the vector space of all periodic signals with period T , and the set of **harmonically related complex exponentials** -

$$\phi_n(t) = e^{jn\omega_0 t}, \quad n \in \mathbb{Z}, \quad \omega_0 = \frac{2\pi}{T}$$

From the way we have defined the inner product, this set of functions $\{\phi_n(t)\}$ is **orthonormal**,

$$\langle \phi_n, \phi_m \rangle = \frac{1}{T} \int_0^T e^{jn\omega_0 t} e^{-jm\omega_0 t} dt = \frac{1}{T} \int_0^T e^{j(n-m)\omega_0 t} dt$$

$$\therefore \langle \phi_n, \phi_m \rangle = \begin{cases} 1, & n = m, & \text{ie, the norm of each basis vector is 1} \\ 0, & n \neq m, & \text{ie, the basis vectors are pairwise orthogonal} \end{cases}$$

Hence $\{\phi_n(t)\}$ is an **orthonormal basis set** - precisely the basis used in the Fourier series expansion.

Fourier Series of Continuous-Time Periodic Signals

Consider a periodic signal $x(t)$ of fundamental period T . We want to represent it in the basis set of **harmonically related complex exponentials**, $e^{jk\omega_0 t}$, where $\omega_0 = 2\pi/T$.

Synthesis equation -

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Analysis equation -

$$a_k = \frac{1}{T} \int_T x(t) e^{-jk\omega_0 t} dt$$

Together, these let us express any periodic signal as a linear combination of harmonically related complex exponentials, with a_k giving the coefficient along each basis vector $\phi_k(t)$.

Existence of Fourier Series

Dirichlet Conditions for existence of the Fourier series of a periodic signal $x(t)$ -

- $x(t)$ is bounded, or is absolutely integrable, ie,

$$\int_T |x(t)| dt < \infty$$

- $x(t)$ has at most a finite number of maxima and minima over one period ("bounded variation")
- $x(t)$ is continuous, or has a finite number of discontinuities, over one period

An alternative condition is **square integrability**, i.e. finite energy over one period -

$$\int_T |x(t)|^2 dt < \infty$$

Convergence of Fourier Series

Consider approximating $x(t)$ over one period by the truncated Fourier series -

$$\hat{x}_N(t) = \sum_{k=-N}^N a_k e^{jk\omega_0 t}$$

The truncated series $\hat{x}_N(t)$ is the **best approximation** of $x(t)$ achievable using only these $2N + 1$ complex exponential terms, because the Fourier coefficients a_k are precisely the ones that **minimize the Mean Squared Error (MSE)** -

$$E_e = \frac{1}{T} \int_T |x(t) - \hat{x}_N(t)|^2 dt$$

As we increase the number of terms N , the MSE, E_e , decreases.

Convergence of Fourier Series

If $x(t)$ satisfies the Dirichlet conditions, then as $N \rightarrow \infty$, the Fourier series converges to $x(t)$ at every point t where $x(t)$ is **continuous**.

At a point of **jump discontinuity**, the Fourier series converges to the midpoint of the right-hand limit (RHL) and left-hand limit (LHL) -

$$\frac{1}{2} [x(t^-) + x(t^+)]$$

Gibbs Phenomenon - Near a jump discontinuity, the truncated Fourier series overshoots/undershoots $x(t)$.

- As $N \rightarrow \infty$, the **size** of this maximum deviation does **not** reduce.
- However, the **width** of the region over which this deviation occurs shrinks to zero.
- Hence, the MSE, E_e , still converges to zero in the limit.

Properties of Fourier Series

For $x(t) \xleftrightarrow{\text{F.S.}} a_k$, $y(t) \xleftrightarrow{\text{F.S.}} b_k$, both periodic with period T , $\omega_0 = 2\pi/T$ -

1) Linearity

$$Ax(t) + By(t) \longleftrightarrow Aa_k + Bb_k$$

2) Time-Reversal

$$x(-t) \longleftrightarrow a_{-k}$$

3) Conjugation

$$x^*(t) \longleftrightarrow a_{-k}^*$$

Properties of Fourier Series

4) Time-Scaling

$$x(\alpha t) \longleftrightarrow a_k$$

(period becomes T/α ; coefficients unchanged, but spectrum changes)

5) Time-Shifting

$$x(t - t_0) \longleftrightarrow e^{-jk\omega_0 t_0} a_k$$

6) Frequency-Shifting

$$e^{jk_0\omega_0 t} x(t) \longleftrightarrow a_{k-k_0}$$

7) Periodic Convolution

$$\int_T x(\tau) y(t - \tau) d\tau \longleftrightarrow T a_k b_k$$

8) Multiplication

$$x(t) y(t) \longleftrightarrow \sum_{n=-\infty}^{\infty} a_n b_{k-n}$$

Fourier Series of Real Signals

For $x(t)$ real, $x(t) = x^*(t)$, which gives **conjugate symmetry** -

$$a_k = a_{-k}^*$$

Equivalently -

$$|a_k| = |a_{-k}| \quad (\text{even magnitude}) \quad \angle a_k = -\angle a_{-k} \quad (\text{odd phase})$$

Writing $a_k = |a_k| e^{j\angle a_k}$, the synthesis equation reduces to -

$$x(t) = a_0 + \sum_{k=1}^{\infty} 2|a_k| \cos(k\omega_0 t + \angle a_k)$$

Fourier Series of Even and Odd Signals

Even-Odd Decomposition of a real signal $x(t)$ -

$$x_{ev}(t) = \frac{1}{2} [x(t) + x(-t)] \qquad x_{od}(t) = \frac{1}{2} [x(t) - x(-t)]$$

Real signals -

$$x(t) \text{ real \& even} \iff a_k \text{ real \& even}$$

$$x(t) \text{ real \& odd} \iff a_k \text{ purely imaginary \& odd}$$

For real $x(t)$, correspondingly -

$$x_{ev}(t) \longleftrightarrow \operatorname{Re}\{a_k\} \qquad x_{od}(t) \longleftrightarrow j \operatorname{Im}\{a_k\}$$

Half-Wave Symmetry

A signal $x(t)$, periodic with period T , is said to have **half-wave symmetry** if -

$$x\left(t - \frac{T}{2}\right) = -x(t)$$

Expanding $x(t - T/2)$ in its Fourier series and comparing coefficients with $-x(t)$ gives -

$$a_k(-1)^k = -a_k$$

which holds only if -

$$a_k = 0 \quad \text{for all even } k$$

That is, a half-wave symmetric signal contains **only odd harmonics** in its Fourier series.

Parseval's Theorem and Bessel's Inequality

Bessel's Inequality -

For a finite-energy signal, the energy contained in any finite number of Fourier coefficients cannot exceed the total signal energy:

$$\sum_{k=-N}^N |a_k|^2 \leq \frac{1}{T} \int_T |x(t)|^2 dt$$

Parseval's Theorem -

The average power of the signal is equal to the sum of the powers contained in its Fourier series coefficients:

$$\frac{1}{T} \int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

Summary of Continuous-time Fourier Series Properties

TABLE 3.1 PROPERTIES OF CONTINUOUS-TIME FOURIER SERIES

Property	Section	Periodic Signal	Fourier Series Coefficients
		$\left. \begin{array}{l} x(t) \\ y(t) \end{array} \right\} \begin{array}{l} \text{Periodic with period } T \text{ and} \\ \text{fundamental frequency } \omega_0 = 2\pi/T \end{array}$	$\begin{array}{l} a_k \\ b_k \end{array}$
Linearity	3.5.1	$Ax(t) + By(t)$	$Aa_k + Bb_k$
Time Shifting	3.5.2	$x(t - t_0)$	$a_k e^{-jk\omega_0 t_0} = a_k e^{-jk(2\pi/T)t_0}$
Frequency Shifting		$e^{jM\omega_0 t} x(t)$	a_{k-M}
Conjugation	3.5.6	$x^*(t)$	a_{-k}^*
Time Reversal	3.5.3	$x(-t)$	a_{-k}
Time Scaling	3.5.4	$x(\alpha t), \alpha > 0$ (periodic with period T/α)	a_k
Periodic Convolution		$\int_T x(\tau)y(t - \tau)d\tau$	$T a_k b_k$
Multiplication	3.5.5	$x(t)y(t)$	$\sum_{l=-\infty}^{+\infty} a_l b_{k-l}$
Differentiation		$\frac{dx(t)}{dt}$	$jk\omega_0 a_k = jk \frac{2\pi}{T} a_k$
Integration		$\int_{-\infty}^t x(\tau) d\tau$ (finite valued and periodic only if $a_0 = 0$)	$\left(\frac{1}{jk\omega_0}\right)a_k = \left(\frac{1}{jk(2\pi/T)}\right)a_k$
Conjugate Symmetry for Real Signals	3.5.6	$x(t)$ real	$\begin{cases} a_k = a_{-k}^* \\ \Re\{a_k\} = \Re\{a_{-k}\} \\ \Im\{a_k\} = -\Im\{a_{-k}\} \\ a_k = a_{-k} \\ \angle a_k = -\angle a_{-k} \end{cases}$
Real and Even Signals	3.5.6	$x(t)$ real and even	a_k real and even
Real and Odd Signals	3.5.6	$x(t)$ real and odd	a_k purely imaginary and odd
Even-Odd Decomposition of Real Signals		$\begin{cases} x_e(t) = \mathcal{E}\{x(t)\} & [x(t) \text{ real}] \\ x_o(t) = \mathcal{O}\{x(t)\} & [x(t) \text{ real}] \end{cases}$	$\begin{cases} \Re\{a_k\} \\ j\Im\{a_k\} \end{cases}$
Parseval's Relation for Periodic Signals			
$\frac{1}{T} \int_T x(t) ^2 dt = \sum_{k=-\infty}^{+\infty} a_k ^2$			

Fourier Series of Discrete-Time Periodic Signals

Consider a periodic signal $x[n]$ of fundamental period N . We want to represent it in the basis set of **harmonically related complex exponentials**, $e^{jk\omega_0 n}$, $\omega_0 = \frac{2\pi}{N}$

Synthesis equation -

$$x[n] = \sum_{k=0}^{N-1} a_k e^{jk\omega_0 n}$$

Analysis equation -

$$a_k = \frac{1}{N} \sum_{n=0}^{N-1} x[n] e^{-jk\omega_0 n}$$

Although the summation can be taken over any N consecutive values of k or n , only N distinct complex exponentials exist.

Periodicity of Discrete Fourier Series Coefficients

The complex exponentials themselves are periodic in the frequency-index k :

$$e^{j(k+N)\frac{2\pi}{N}n} = e^{jk\frac{2\pi}{N}n} e^{j2\pi n} = e^{jk\frac{2\pi}{N}n}.$$

Hence,

$$a_{k+N} = a_k$$

Therefore, unlike the continuous-time Fourier Series, the sequence of Fourier coefficients $\{a_k\}$ is itself **periodic with period N** .

Thus there are only N **distinct Fourier coefficients**.

Properties of Discrete Fourier Series

For $x[n] \longleftrightarrow a_k$, $y[n] \longleftrightarrow b_k$, both periodic with period N , $\omega_0 = 2\pi/N$,

1) Linearity

$$Ax[n] + By[n] \longleftrightarrow Aa_k + Bb_k$$

2) Time-Reversal

$$x[-n] \longleftrightarrow a_{-k}$$

3) Conjugation

$$x^*[n] \longleftrightarrow a_{-k}^*$$

4) Time-Shifting

$$x[n - n_0] \longleftrightarrow e^{-jk\omega_0 n_0} a_k$$

5) Frequency-Shifting

$$e^{jk_0\omega_0 n} x[n] \longleftrightarrow a_{k-k_0}$$

Properties of Discrete Fourier Series

7) Periodic Convolution

$$z[n] = \sum_{r=0}^{N-1} x[r] y[n-r]$$

$$z[n] \longleftrightarrow N a_k b_k$$

8) Multiplication

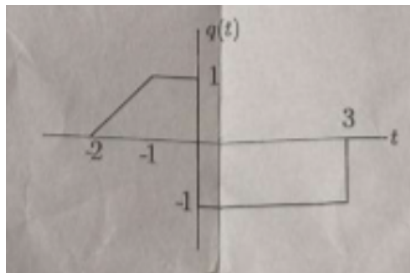
$$x[n] y[n] \longleftrightarrow \sum_{r=0}^{N-1} a_r b_{k-r}$$

9) Parseval's Relation

$$\frac{1}{N} \sum_{n=0}^{N-1} |x[n]|^2 = \sum_{k=0}^{N-1} |a_k|^2$$

Question

Given the signal $q(t)$ in the figure below, sketch the transformed signal $p(t) = q(-1 - 2t)$. Show intermediate stages for partial marks.



Question

State with justification, whether the given system is invertible -

(a) $y(t) = x(2t)$

(b) $y[n] = x[2n]$

Question

Indicate whether the given system is (i) memoryless, (ii) causal, (iii) linear, and (iv) time-invariant. Justify your answers.

$$y[n] = x[n] + nx[n + 1]$$

Question

Determine whether the following systems are (i) linear, (ii) time-invariant,

(a) $y(t) = g(t)x(t)$

(b) $y(t) = x(t/3)$

Question

Perform the following continuous-time convolution:

$$(u(t) - u(t - 2)) * u(t).$$

Sketch the output signal.

Question

Perform the following discrete-time convolution:

$$(\delta[n] + \delta[n - 1]) * (\delta[n] - \delta[n - 2]).$$

Sketch the output signal.

Question

State with justification whether the given LTI system impulse response corresponds to a causal and/or stable system.

(a) $h(t) = e^{-2t}u(t + 50)$

(b) $h(t) = te^{-t}u(t)$

Question

- (a) Show that, if the input to a time-invariant system is periodic, the output is periodic with the same period T .
- (b) Given a finite duration input signal, give an example of an LTI system impulse response such that the output is periodic with fundamental frequency ω_0 .

Question

For each of the below, state the expected region of zero values in the output of the convolution $x(t) * h(t)$ in terms of t or n as required.

(a) $x(t) = 0$ for $t > T_1$ and $h(t) = 0$ for $t > T_2$.

(b) $x[n] = 0$ for $n < N_1$ and $h[n] = 0$ for $n < N_2$.

Question

Given an LTI system with impulse response $h(t) = e^{2t}u(-t)$, find and sketch the output $y(t)$ for the input $x(t) = u(t) - 2u(t - 2) + u(t - 5)$.

Question

For the continuous-time periodic signal

$$x(t) = 2 + \cos\left(\frac{2\pi}{3}t\right) + 4\sin\left(\frac{5\pi}{3}t\right)$$

determine the fundamental frequency ω_0 and the Fourier Series coefficients.

Question

Let $x(t)$ be periodic with period T , and

$$x(t) \longleftrightarrow \{a_k\}$$

Find the Fourier coefficients of $x(3t - 1)$.

Question

Given the Fourier series coefficients of a periodic signal $x(t)$ are as follows,

$$a_k = \begin{cases} 2, & k = 0 \\ j \left(\frac{1}{2}\right)^{|k|}, & \text{otherwise} \end{cases}$$

use Fourier series properties to answer each of the following questions by first stating and deriving the required property.

- (a) Is $x(t)$ real?
- (b) Is $x(t)$ even? (i.e. $x(t) = x(-t)$)
- (c) Is $\frac{dx(t)}{dt}$ even?

Question

Consider a periodic signal $x(t)$ whose Fourier-series coefficients are given as

$$a_k = j \left(\frac{1}{2} \right)^{|k|}$$

Let the energy of the signal $x(t)$ over one time period be equal to 5. Determine the time period of the given signal.

Question

Consider a discrete-time periodic real-valued sequence $x[n]$ with period $N = 5$. The Fourier series representation of the sequence is given as $x[n] = \sum_{k=0}^4 a_k e^{j\frac{2\pi}{5}kn}$, where $a_0 = 1$, $a_1 = 2j$, and $a_2 = 3j$. Find all a_k needed to fully specify $x[n]$.

The sequence is passed through an LTI filter with impulse response $h[n] = \delta[n] - \alpha\delta[n - 2]$ to get the output sequence $y[n]$. Find

$$\sum_{n=0}^4 y[n] \sin\left(\frac{4\pi n}{5}\right).$$

What is the numerical value of the sum when the scaling factor is set to $\alpha = \frac{1}{\cos(2\pi/5)}$?